

Introduction to Machine Learning - Decision Tree Coursework

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1 Evaluation before pruning

1.1 Cross Validation Classification Metrics

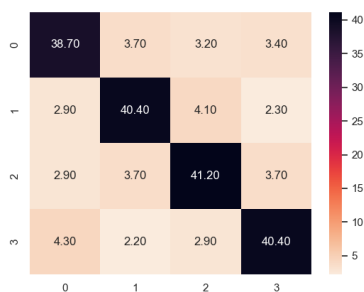
1.1.1 Clean Dataset - Confusion Matrix & Metrics



Accuracy = 97.34%

	Class 1	Class 2	Class 3	Class 4
Precision	0.988	0.964	0.951	0.991
Recall	0.992	0.959	0.950	0.991
F1	0.990	0.961	0.950	0.991

1.1.2 Noisy Dataset - Confusion Matrix & Metrics



Accuracy = 80.35%

	Class 1	Class 2	Class 3	Class 4
Precision	0.790	0.808	0.802	0.807
Recall	0.791	0.813	0.803	0.811
F1	0.787	0.808	0.801	0.807

1.2 Result Analysis

For the clean dataset, the decision tree classifies rooms 1 and 4 highly accurately. Occasionally rooms 2 and 3 are confused with each other, but these rooms are also classified highly accurate overall. The average accuracy for the decision trees trained on clean dataset was 97.34%. For the noisy dataset, rooms are still classified fairly accurate, but there are more confusions between rooms, especially between adjacent rooms. The average accuracy for the decision trees trained on noisy dataset was 80.35%.

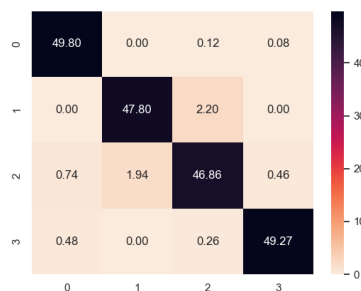
1.3 Dataset Differences

From the data above, we observe that the tree trained on clean dataset achieved significantly higher results than the one trained on noisy data. The reason for the difference in metrics is due to the noise in the second dataset. The noisy data contains close, floating point, values, making classification harder than in the clean data, where the differences in values are larger. Trees trained on noisy data are larger, because of additional branches created to fit the noisy data, causing overfitting. As a result, these perform worse on unseen data.

2 Evaluation after pruning

2.1 Cross Validation Classification Metrics

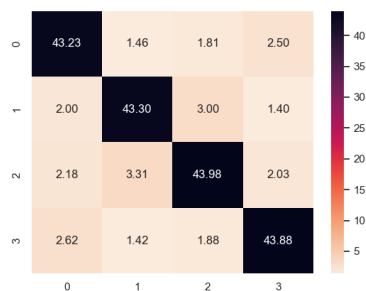
2.1.1 Clean Dataset - Confusion Matrix & Metrics



Accuracy = 96.86%

	Class 1	Class 2	Class 3	Class 4
Precision	0.976	0.961	0.948	0.988
Recall	0.996	0.955	0.936	0.985
F1	0.986	0.958	0.942	0.986

2.1.2 Noisy Dataset - Confusion Matrix & Metrics



Accuracy = 87.19%

	Class 1	Class 2	Class 3	Class 4
Precision	0.862	0.870	0.868	0.878
Recall	0.882	0.869	0.853	0.881
F1	0.870	0.868	0.860	0.878

2.2 Result Analysis after Pruning

On the clean dataset the pruning did not have a big impact on performance, as the decision trees before pruning did not overfit, unlike the ones obtained for the noisy data. As we can see in the tables above, pruning had a big effect on the noisy dataset, increasing accuracy by about 7% and lowering the average depth. We can see big differences in metrics, especially in the precision and recall of classes 2 and 3, which were previously confused often. Moreover, all pairs of labels are misclassified less than before.

2.3 Depth Analysis

	Clean Dataset		Noisy Dataset	
	Unpruned	Pruned	Unpruned	Pruned
Depth	12.2	8.64	18.7	13.86
Accuracy	97.34%	96.86%	80.35%	87.19%

Average depth and accuracy after nested cross validation

The average depth after pruning drops about 28% for the two datasets, which leads to increased accuracy for noisy data and no significant changes for clean data. Assuming the clean dataset has no noise, training on this data can't lead to overfitting, so pruning will not improve performance, and may even lower it. For noisy data, pruning helps the tree extract a general pattern, reducing the overfitting. We notice that depth, accuracy and overfitting are closely related as in the case of overfitting data, reducing the depth increases accuracy.

3 Tree visualisation

3.1 Decision Tree trained on 100% of clean dataset

